

Business Requirements Document (BRD)

Multi-Acquirer Payment Gateway

Document Version: 1.0

Date: 2026-06-08

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Status: Approved

1. Executive Summary

This document outlines the business requirements for a **Multi-Acquirer Payment Gateway** — a centralized payment orchestration platform that enables merchants to process payments across multiple acquiring banks and payment processors through a single, unified API. The system provides intelligent routing, failover, cost optimization, and PCI-DSS compliant transaction management.

2. Business Context

2.1 Problem Statement

Merchants currently face the following challenges:

- **Single point of failure:** Reliance on one acquirer leads to revenue loss during outages.
- **High transaction costs:** No ability to route to the lowest-cost acquirer dynamically.
- **Geographic limitations:** Single acquirers lack global coverage, reducing authorization rates in some regions.
- **Operational complexity:** Integration with multiple acquirers requires separate codebases and contracts.
- **Compliance overhead:** Each acquirer integration must independently meet PCI-DSS standards.

2.2 Business Opportunity

Metric	Current State	Target State
Payment success rate	91%	97%+
Average transaction cost	\$0.35	\$0.22
Integration time per acquirer	8 weeks	1 week
Acquirers supported	1	5+
Uptime SLA	99.5%	99.99%

3. Stakeholders

Role	Name/Team	Interest
Product Owner	Payments Team	Feature prioritization
Merchant Operations	Business Dev	Onboarding, SLA
Finance	CFO Office	Cost reduction
Security & Compliance	InfoSec	PCI-DSS, fraud
Engineering	Platform Team	Technical implementation
Acquirer Partners	Stripe, Adyen, Worldpay, Braintree	Integration standards

4. Business Requirements

4.1 Core Payment Processing

BR-001 — The system SHALL accept payment authorization requests via a RESTful API from authenticated merchants.

BR-002 — The system SHALL support the following payment types: - Credit card (Visa, Mastercard, Amex, Discover) - Debit card - Digital wallets (Apple Pay, Google Pay) - ACH / bank transfers

BR-003 — The system SHALL support the following transaction types: - Authorization - Capture (same-day and deferred) - Authorization + Capture (sale) - Void / Reversal - Refund (full and partial)

BR-004 — The system SHALL process payment authorization within **2 seconds** (p95 latency) under normal load.

BR-005 — The system SHALL support amounts in multiple currencies (ISO 4217) and handle FX conversion metadata.

4.2 Multi-Acquirer Routing

BR-010 — The system SHALL maintain connections to a minimum of **3 active acquirers** simultaneously.

BR-011 — The system SHALL route transactions based on configurable rules including: - Card network preference (e.g., Visa transactions → preferred acquirer) - Geographic region of issuer - Transaction amount thresholds - Merchant category code (MCC) - Merchant-specific acquirer preferences

BR-012 — The system SHALL support **cost-based routing** to minimize interchange and acquirer fees.

BR-013 — The system SHALL support **performance-based routing** using real-time success rate metrics per acquirer.

BR-014 — The system SHALL implement **automatic failover**: if the primary acquirer returns a network or system error, the transaction SHALL be retried with a secondary acquirer within **500ms**.

BR-015 — Failover SHALL NOT occur for hard declines (insufficient funds, card stolen) — only for acquirer system errors.

4.3 Security & Compliance

BR-020 — The system SHALL be **PCI-DSS Level 1** compliant.

BR-021 — The system SHALL **never store raw PAN (Primary Account Number)**. All card data SHALL be tokenized before persistence.

BR-022 — All data in transit SHALL be encrypted using **TLS 1.2+**.

BR-023 — All stored sensitive data SHALL be encrypted using **AES-256**.

BR-024 — The system SHALL implement **3DS2 (3D Secure 2.0)** authentication support.

BR-025 — API access SHALL require **OAuth 2.0 / API key** authentication with per-merchant scoped permissions.

BR-026 — The system SHALL perform basic **fraud scoring** on each transaction and block transactions exceeding configurable thresholds.

4.4 Idempotency & Reliability

BR-030 — The system SHALL support **idempotency keys** to prevent duplicate charges on network retries.

BR-031 — The system SHALL guarantee **exactly-once** payment processing semantics for any given idempotency key within a 24-hour window.

BR-032 — The system SHALL maintain a complete, immutable **audit log** of all transaction state changes.

BR-033 — The system SHALL achieve **99.99% uptime** (SLA), permitting no more than 52 minutes downtime per year.

4.5 Merchant Management

BR-040 — The system SHALL support multi-tenant merchant onboarding with isolated data and credentials.

BR-041 — Each merchant SHALL have configurable routing profiles, acquirer assignments, and fee structures.

BR-042 — The system SHALL expose a management API for merchants to query transaction status, request refunds, and view settlement reports.

4.6 Reporting & Analytics

BR-050 — The system SHALL provide real-time transaction status via API.

BR-051 — The system SHALL expose aggregated metrics: success rate, decline rate, volume, and latency per acquirer, per merchant, per time window.

BR-052 — The system SHALL generate daily settlement reconciliation files in ISO 8583 / CSV format.

5. Non-Functional Requirements

Category	Requirement
Throughput	1,000 TPS sustained; 5,000 TPS burst
Latency	p95 < 2s end-to-end; p99 < 4s
Availability	99.99% (multi-region active-active)
Data Retention	Transaction records: 7 years (regulatory)
Scalability	Horizontal scale with zero downtime
Observability	Distributed tracing, metrics, structured logs
Disaster Recovery	RTO < 15 min; RPO < 1 min

6. Assumptions & Constraints

- Acquirer integrations will use HTTP/REST or SOAP APIs (no direct ISO 8583 host-to-host initially).
 - Card data entry happens client-side (merchant's frontend) using tokenization SDKs; raw card data never reaches this service.
 - Initial deployment on AWS (EKS); multi-cloud is out of scope for Phase 1.
 - Regulatory compliance (PCI-DSS audit, penetration testing) is a separate track but the system must be designed for it.
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7. Out of Scope (Phase 1)

- Payment links / hosted payment pages (future Phase 2)
 - Subscription / recurring billing engine
 - Direct ISO 8583 host-to-host acquirer connections
 - Marketplace / split payments
 - Crypto payments
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8. Success Criteria

KPI	Target	Measurement
Payment success rate	$\geq 97\%$	Monthly avg
P95 authorization latency	≤ 2 seconds	Real-time monitoring
System availability	$\geq 99.99\%$	Uptime monitoring
Acquirer failover success	$\geq 99\%$ of failover attempts	Log analysis
Cost per transaction	$\leq \$0.22$	Finance reports

9. Timeline (Phased Delivery)

Phase	Scope	Duration
Phase 1 – MVP	Core API, 2 acquirers (Stripe + Adyen), basic routing, idempotency	Sprint 1-4 (8 weeks)
Phase 2 – Routing	Smart routing, failover, cost-based rules, 2 more acquirers	Sprint 5-8 (8 weeks)
Phase 3 – Scale	3DS2, fraud scoring, analytics dashboard, EKS autoscaling	Sprint 9-12 (8 weeks)
Phase 4 – Enterprise	Multi-region, settlement reports, merchant management UI	Sprint 13-16 (8 weeks)

End of BRD